Cambridge Assessment International Education

Cambridge O Level

PHYSICS **5054/22**

Paper 2 Theory **May/June 2023**

MARK SCHEME

Maximum Mark: 75

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

+¬?-

© UCLES 2023

[Turn over

PUBLISHED

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

© UCLES 2023

GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

© UCLES 2023

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	'List rule' guidance For questions that require ***n*** responses (e.g. State **two** reasons): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

© UCLES 2023

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

© UCLES 2023

Question	Answer	Marks
1(a)	speed is constant for 10 s / at first	**B1**
	deceleration • has (average) value 0.39 m / s^2 • lasts for 40 s • is non-uniform	**B1**
1(b)(i)	friction (with road) **and** air resistance **or** (air) drag	**B1**
1(b)(ii)	resistive force(s) change / decrease (as speed decreases) **or** (graph shows) deceleration is not constant / decreases **or** non-uniform decrease in speed	**B1**
1(c)(i)	(momentum =) *mv* in any form algebraic or numerical	**C1**
	8400 **and** kg m / s **or** Ns	**A1**
1(c)(ii)	(F =) $\Delta mv / t$ **or** $F = ma$ in any form algebraic or numerical	**C1**
	840 (N)	**A1**

Question	Answer	Marks
2(a)(i)	chemical (potential energy)	**B1**
2(a)(ii)	electrical work / (electrical) current **or** electrons (made to) move	**B1**
	mechanical) work is done by the motor / by or on the (back) wheel	**B1**
2(b)	(KE =) ½ mv^2 in any form	**C1**
	560 (J)	**A1**
2(c)(i)	(energy provided if power is) 1 kW for 1 hour / 1000 W for 1 hour	**B1**

© UCLES 2023

Question	Answer	Marks
2(c)(ii)	*(t =) E / P* **or** 0.35 / 0.07 **or** 350 / 70	**C1**
	5(.0 hours)	**A1**

Question	Answer	Marks
3(a)(i)	mercury exerts a changing or reduced pressure / force (when vertical but not when horizontal) / mercury stops exerting a pressure	**B1**
3(a)(ii)	particles hit walls / mercury	**B1**
	fewer hits per second **or** hit less often / less frequently **or** fewer hits per unit area	**B1**
	less force (per unit area on wall)	**B1**
3(b)	(*P*=) *dgh* in any form	**C1**
	41 000 seen	**C1**
	140 000 (Pa)	**A1**
3(c)	curved shape with decreasing gradient	**B1**
	points plotted at or curve through ($½V_0$, $2P_0$), (P_0, V_0) **and** ($2V_0$, $½P_0$)	**B1**

© UCLES 2023

Question	Answer	Marks
4(a)(i)	–273 (°C) **and** 0 (K)	**B1**
4(a)(ii)	particles) move fast(er)	**B1**
	vibrate / oscillate with larger amplitude	**B1**
4(a)(iii)	internal energy increases	**B1**
	temperature stays the same	**B1**
4(b)(i)	(*E* =) *mcT* in any form numerical or algebraic	**C1**
	8000 (J)	**A1**
4(b)(ii)	100–44 **or** 56	**C1**
	0.48 **or** 0.47 (J / (g °C))	**A1**

Question	Answer	Marks
5(a)	seismic S-waves **and** X-rays	**B1**
5(b)(i)	(wavelength =) *v* / *f* in any form **or** (distance =) *vt* **or** *f* = 1 / *T* or 1 / 0.15	**C1**
	frequency 6.7 (Hz)	**A1**
	wavelength 4.0 (cm)	**A1**
5(b)(ii)	straight wavefronts above block to the right with same wavelength	**B1**
	correct diffraction – wavefronts quarter circles with centre at top corner	**B1**
5(b)(iii)	increase frequency of movement of (wooden) bar	**B1**

© UCLES 2023

Question	Answer	Marks
5(b)(iv)	less diffraction **or** less bending	**B1**

Question	Answer	Marks
6(a)(i)	ammeter A_2 and A_3 both 0.25 (A)	**B1**
	voltmeter V_2 9(.0 V)	**B1**
6(a)(ii)	(R =) V / I in any form	**C1**
	36 (Ω)	**A1**
6(a)(iii)	current is (directly) proportional to voltage / potential difference	**B1**
	provided the temperature is constant	**B1**
6(b)	0.35 (A) seen **or** total resistance 36 + 6 / 0.35; 36 + 17(.1); 53(.1) Ω	**B1**
	voltage (across R) 0.35 × 36 **or** 12.6 (V) seen	**B1**
	19 V	**B1**

Question	Answer	Marks
7(a)(i)	X-rays **and** gamma (radiation / rays)	**B1**
7(a)(ii)	security marking / detecting counterfeit bank notes / sterilising water / fluorescent effects (e.g. fingerprints, blood, biological fluids) / (sun) tanning / sun beds / cancer treatment / curing inks and resins (e.g. in dental fillings) / production of vitamin (D) / generating electricity / solar panels / disinfection / sterilisation	**B1**
7(a)(iii)	(skin) cancer / cataracts / burns or destroys the retina or cornea	**B1**

© UCLES 2023

Question	Answer	Marks
7(b)(i)	critical angle	**B1**
7(b)(ii)	(*n* =) sin*i* / sin*r* **or** 1 / sin*c*	**C1**
	1.6	**A1**

Question	Answer	Marks
8(a)	(soft) iron	**B1**
8(b)	input voltage or input current is alternating	**B1**
	changing (magnetic) field (in primary causes changing magnetic field in secondary) (alternating emf is) induced (in secondary coil)	**B1**
	or (magnetic) field lines cut secondary	**B1**
8(c)	(*N*s=) 4600 × 5 / 230 **or** *V*p / *V*s = *N*p / *N*s in any form	**C1**
	100	**A1**
8(d)	reverse the coils **OR** connect the input to the secondary / right-side coil **OR** secondary has more turns than primary / more than 4600 (with same number on primary)	**B1**

Question	Answer	Marks
9(a)(i)	alpha	**B1**
9(a)(ii)	beta	**B1**
9(b)(i)	beta particles because count rate goes down (by 600) when metal placed in position	**B1**

© UCLES 2023

Question	Answer	Marks
9(b)(i)	beta absorbed by metal / cannot pass through metal	**B1**
	gamma (because) radiation passes through metal / still gives reading with thick metal	**B1**
9(b)(ii)	alpha stopped by a few cm in air / alpha do not reach G.M. tube **or** alpha not detected by G.M. tube (end cap stops them)	**B1**
9(c)	use tongs / handle at distance / place absorber between source and observer / limit exposure time / place in lead box or thick metal box	**B1**

Question	Answer	Marks
10(a)(i)	distance travelled (in a vacuum) by light in one year	**B1**
10(a)(ii)	100 000	**B1**
10(b)(i)	supernova	**B1**
	nebula **and** neutron star	**B1**
10(b)(ii)	in a supernova	**B1**

© UCLES 2023

Question	Answer	Marks
10(c)	**either** red shift / increase in (observed) wavelength / reduction in frequency mentioned **or** galaxies / stars moving away (from the Earth) / separating	**B1**
	further away (the galaxy / star) the greater its speed / greater red shift / greater increase in wavelength	**B1**
	going backwards in time / at start) stars or galaxies were close together **or** high density / dense	**B1**
	cosmic microwave background mentioned **or** universe expands	B1
	remnant heat / radiation / left over radiation / radiation from early universe observed now	B1
	radiation has red shifted / longer wavelengths / smaller frequency / become cooler **or** CMBR is uniform / observed in all directions	B1

© UCLES 2023